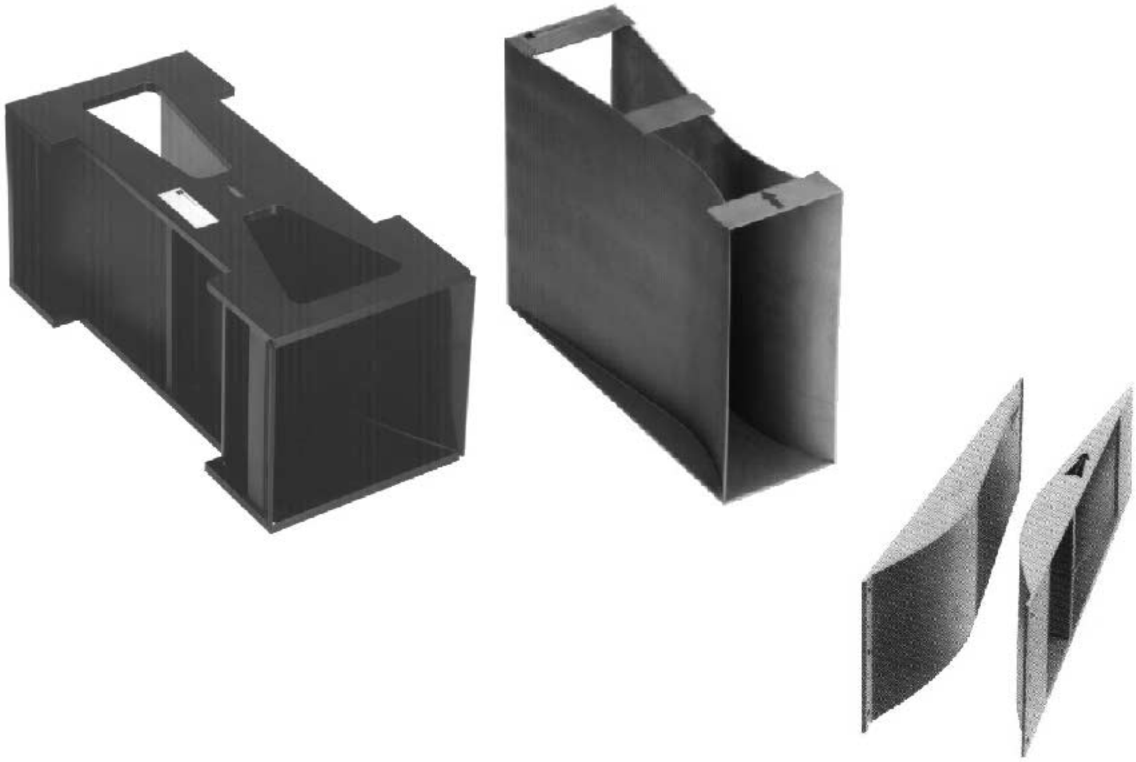
Technical Information
TI003/11/en

Flow Measurement

Khafagi-Venturi QV 302...QV 316

Open channels for flow measurement with ultrasonics



Features and Benefits

- Nine standard sizes for flowrates from 0.4 l/s to 1500 l/s
- Special designs as standard - for adapting to site conditions
- High accuracy with version-specific calibration
- Available in toughened plastic or chrome/nickel steel
- Resistant to acidic or alkaline wastewater, with no washout and low build-up
- Low upper water level using a flow-optimised measuring channel

Applications

Khafagi-Venturi flumes are used for measuring the outflow of open channels such as the inflow and outflow of industrial and communal wastewater. Khafagi-Venturi flumes are best installed when directly fitted into a new channel under construction. It is a fully calibrated unit which guarantees maximum accuracy. Khafagi-Venturi semicoques can also mounted in channels at low cost at a later date.

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The Measuring System

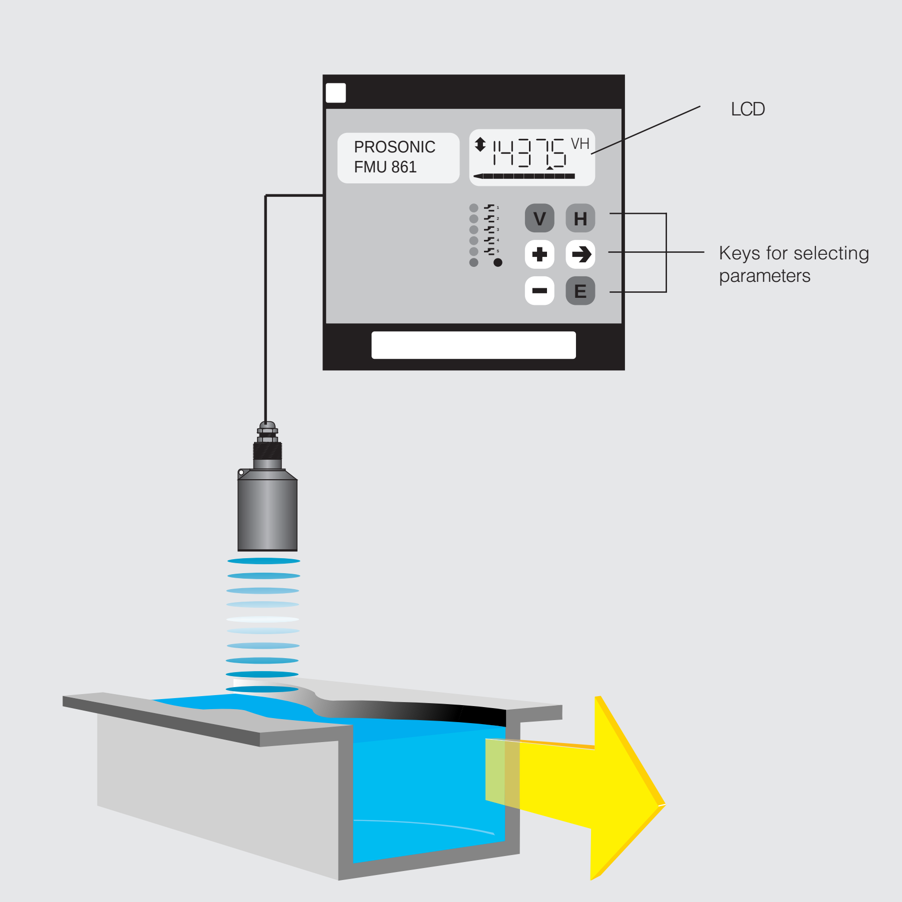
The Khafagi-Venturi are used in an open channel and provide a direct relationship between flowrate (l/s...m³/h) and the upper water level. The flowrate can thus be determined from the height of the water directly upstream of the Khafagi-Venturi constriction.

The level of the headwater is measured by a non-contact, and thus maintenance-free, ultrasonic level measuring system, e.g. Prosonic. An integrated flow computer then converts the measured value into a flowrate at the outflow.

The amount of water flowing is totalised and shown by an internal counter (see Fig.).

Additional features:
- Creep suppression
- Sampling as a function of either quantity or time

The linearisation curve of all common standard channels and weirs are already programmed in and can be called up. Special constructions can be individually programmed as required.



Khafagi-Venturi flume

Construction

The inlet to the throat is an arc of a circle. Because the length of the inlet matches that of the channel, friction is insignificant with small effects caused by the curvature of stream threads. A larger outflow is therefore achieved to give the same upstream level in comparison with other measuring channels.
For Khafagi-Venturi flumes, the value was selected for the ratio b2/b1 (throat width/inlet width) for an optimum value between the water height at the inlet and accuracy of the system.
Widening after the throat (= diffuser) is in the ratio of 1:8 in order to keep losses as low as possible. A change in flow is produced in the throat creating a difference in height which is required for the measuring at the outflow.
The decisive advantage over weirs is that the flume has a continuous, flat and smooth base to ensure that no sediment accumulates upstream or in the flume at certain flow velocities. This guarantees improved long-term accuracy without the need for maintenance.

The Khafagi-Venturi flumes are individually tested on a work bench at the Institute for Hydraulic Engineering, University of Stuttgart using the formula below for the outflow:

$Q \triangleq 0{,}01744 \bullet b^2 \bullet h^{1,5} + 0{,}00091 \bullet h^{2,5}$

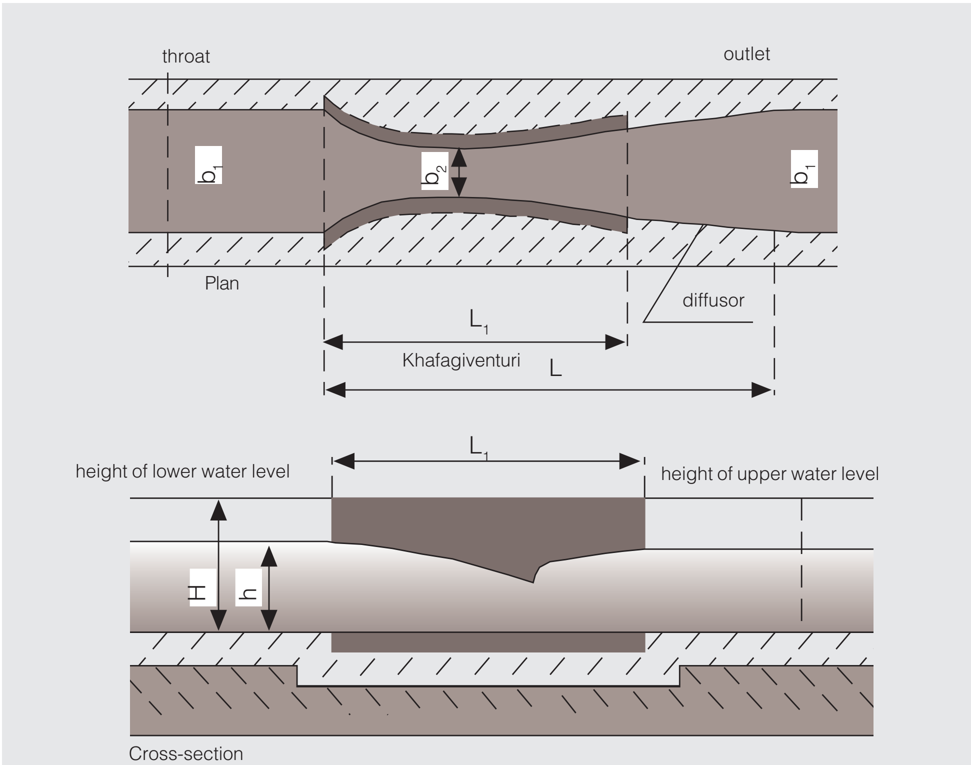
$Q \triangleq$ outflow [l/s]

$b_2 \triangleq$ throat width [cm]

$h \triangleq$ height upstream of the flume [cm]

When calibrated, the Khafagi-Venturi flumes have a variation of maximum 2% over the outflow range from 6% to 20%. This variation is a maximum 1% over the outflow range of 20% to 100%.
The variation of final reading is below 0.4% over the entire range of outflow.

The Khafagi-Venturi Flumes have a long operational life and barely require maintenance due to the use of materials which have high resistance to chemicals and mechanical wear. Depending on the quality of the water, the flumes can be made of a range of materials which significantly increase their operating life.



Dimensions of Khafagi-Venturi flumes

b1 = throat width
b2 = constriction width
L1 = flume length
L = length up to end of diffuser
H = total height of flume
h = height of upper water level

Flowrates of standard sizes for compete Khafagi-Venturi flumes
(standard versions have higher side walls)

Type	for channel width b1	Maximum flowrate Q		with higher side walls		height of upper water level h with Q_{max}	with higher side walls
	mm	l/s	m³/h	l/s	m³/h	mm	mm
QV 302	120	11	40	22	80	224	324
QV 303	300	25	90	50	180	228	351
QV 304	400	50	180	100	360	297	461
QV 305	500	90	320	180	640	381	585
QV 306	600	100	360	200	720	366	567
QV 308	800	250	900	500	1800	557	853
QV 310	1000	500	1800	1000	3600	752	1158
QV 313	1300	800	2880	1600	5760	870	1343
QV 316	1600	1500	5400	3000	10800	1147	1768

Planning and Installation for Khafagi-Venturi Flumes

Install the measuring channel at a place where the outlet leading to the Khafagi-Venturi flume is a gentle flow and not agitated, i.e. with normal flow profile. Any dropped beds creating a water surge or bends immediately upstream from the measuring system can lead to large errors when measuring the quantity of water. There should thus be a straight channel path of at least 10 x b upstream from the measuring system (b1 = channel width)

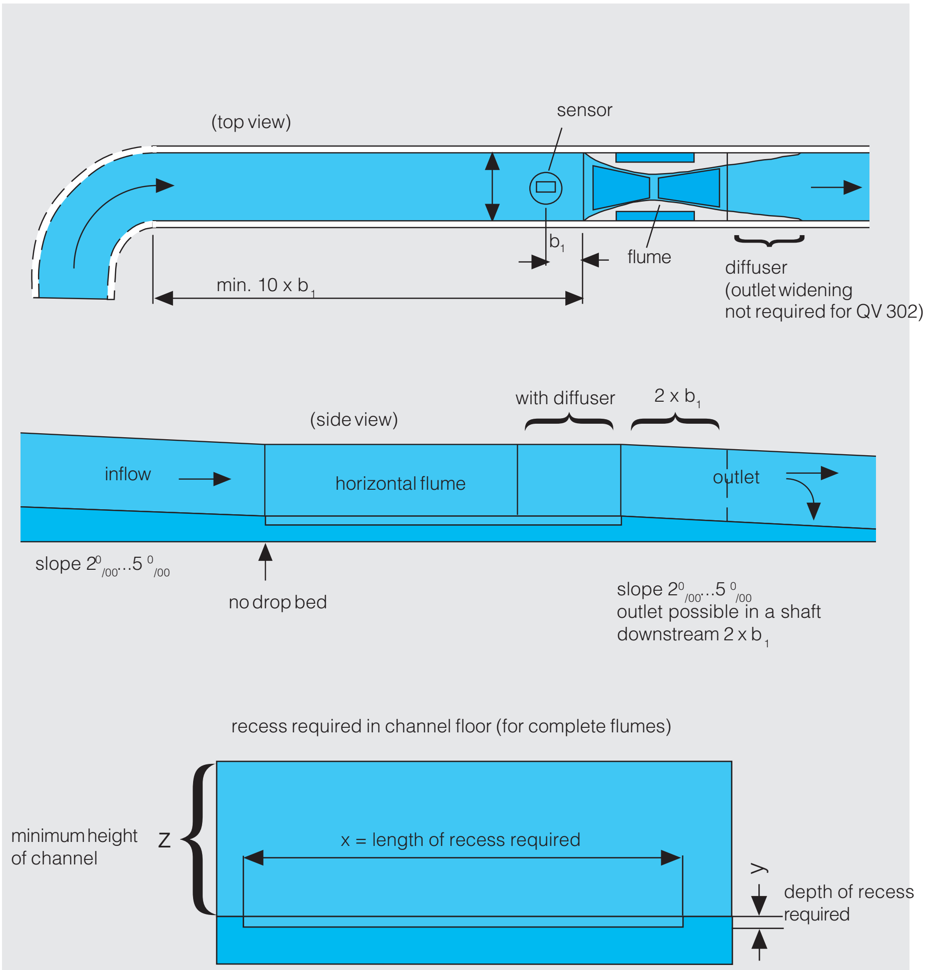
The following channel paths of gentle flow should be used:

a) 10 x b1, downstream from a bend
b) 30 x b1, downstream from lateral flow
c) 50 x b1, downstream from a weir

For flumes fed by a pipe, then a straight-planed section 3 x b1 upstream from the flume is sufficient once the profile has changed from rounded to straight-planed. All other requirements are the same as for a, b and c. The slope of the channel should be approx. $2^{0}{}_{/00}...5\ ^{0}{}_{/00}$. Walls and base of the channel should be as possible. The flow velocity of water should be ≥ 0.6 m/s so that the solids content is easily carried away. The slope should, however, not be too large as otherwise the outlet downstream from the Khafagi-Venturi will gush out too quickly. There should be no mechanical parts in the outlet flume to affect the measuring height. The longitudinal axis of the flume must exactly match that of the inlet flume. The Khafagi-Venturi flume can be positioned accurately using the four reference on the upper surface of the flume.

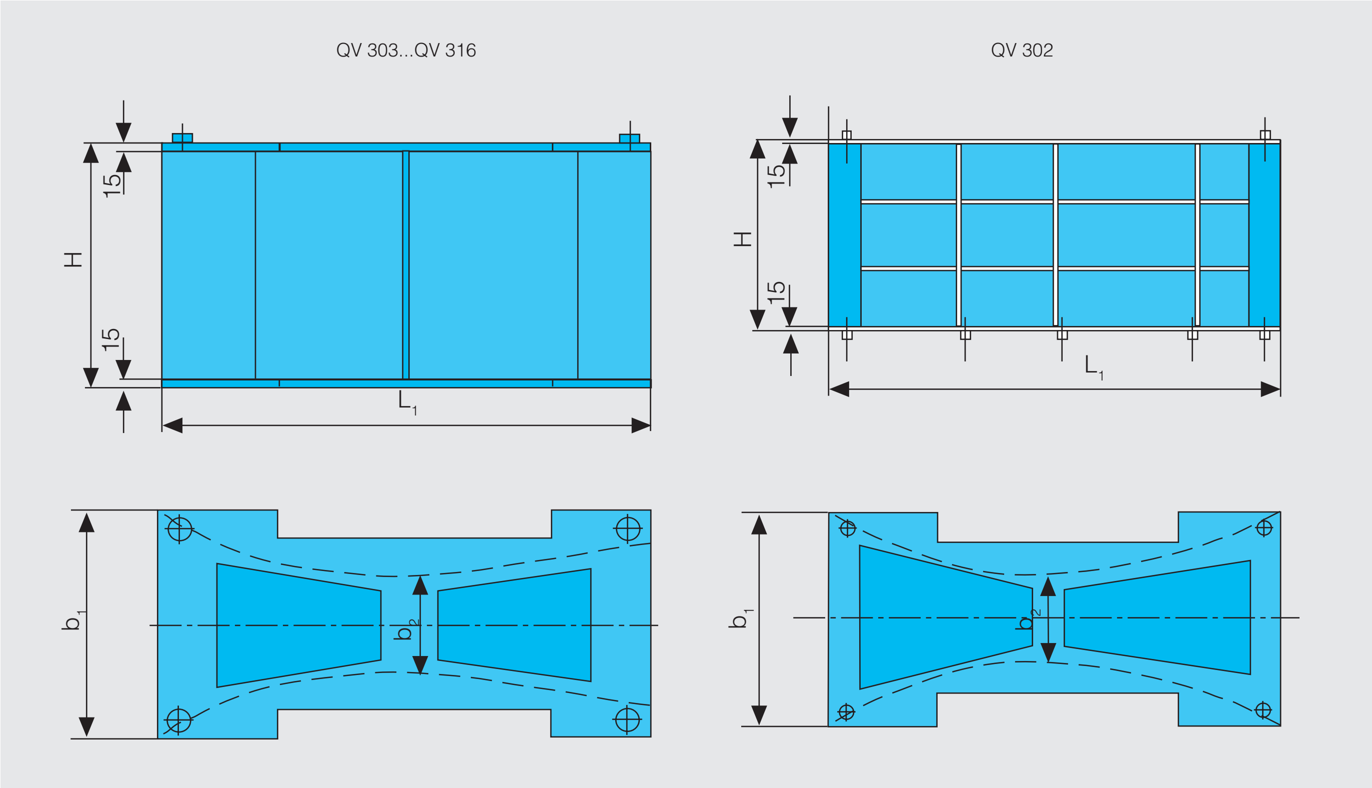
Prepare the cement base and set the flume on it while ensuring that:
- the channel is in the correct direction of flow
- the floor is exactly horizontal
- there is no drop bed at the inlet
- the flume is flush with the channel
- no water remains in the measuring channel when the feed channel is dry

There should be expansion gaps of 10...15 mm at the inlet and at the outlet of the flume and filled with a permanently elastic grouting.
The channel next to the side walls is filled with a non-compressible lean concrete (do not vibrate!).



Technical Data Khafagi-Venturi Flumes

The QV 302 Khafagi-Venturi flume is supplied with a diffuser outlet.
Flumes QV 303 to QV 316 are supplied without a diffuser.
Khafagi-Venturi flumes with raised side walls enable the flowrate to be doubled while the channel width b1 remains the same.
Sizes between standard widths are also available as options.



Dimensions of Khafagi-Venturi
QU302..QV316

Standard sizes (all dimensions in mm)
for complete Khafagi-Venturi flumes
Constriction ratio b_2:b_1 = 0.4):

Type	QV 302	QV 303	QV 304	QV 305	QV 306	QV 308	QV 310	QV 313	QV 316
Throat width b1	120	300	400	500	600	800	1000	1300	1600
Constriction width b2	48	120	160	200	240	320	400	520	640
Flume length L1	503	690	920	1150	1380	1840	2300	3000	3680
Length, up to end of diffuser L	503	1050	1400	1750	2100	2800	3500	4550	5600
Total height of flume H	300	300	400	450	450	670	870	1020	1320
Total height with raised side walls H	400	400	500	600	650	870	1200	1400	1800
Length of recess X*	520	710	940	1170	1400	1860	2330	3030	3710
Depth of recess Y*	40	40	40	40	40	60	60	60	60
Min. height of channel wall Z*	285	285	385	435	435	655	855	1005	1305
Min. height of channel wall with raised side walls	385	385	485	585	635	855	1185	1385	1785

* see Fig.
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Planning and mounting instrumentation for Khafagi-Venturi halfshells

Note the following points for mounting two half-shells in an existing channel instead of mounting a complete flume

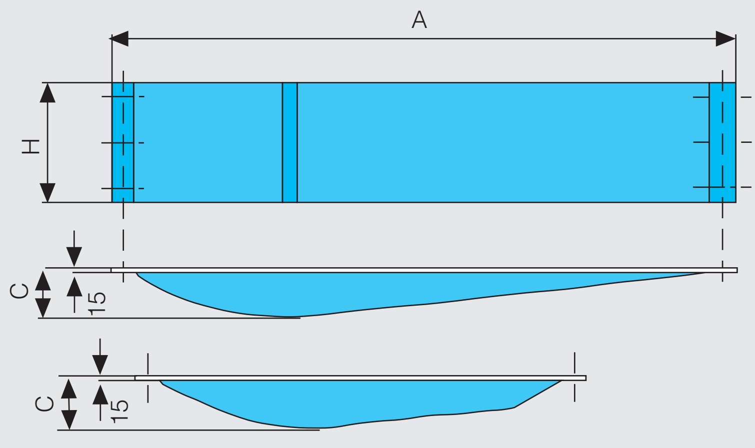
- The same paths are used for normal flow as for complete Khafagi-Venturi flumes
- Ensure that the installation point has a channel bottom which is even and smooth.
- The half-shells must be exactly opposite one another.
- The half-shells must lie on the base of the channel
- The size b2 (throat width) and b1 (inlet width = outlet width) must be constant - from the upper edge of the lower edge of the half-shells.

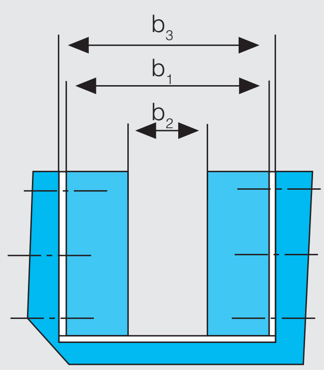
There should be no edges on the side walls nor on the path between the flume and the diffuser. Allowance should be made for the drop bed.

Technical Data Khafagi-Venturi halfshells

Standard sizes (all dimensions in mm) for complete Khafagi-Venturi flumes Constriction ratio b_2:b_1 = 0.4):

Type	QV 302	QV 303	QV 304	QV 305	QV 306	QV 308	QV 310	QV 313	QV 316
Flume length A	600	1250	1600	1950	2300	3050	3200	4000	4800
Height H	300	300	400	450	450	670	870	1020	1320
Height with raised side walls H1	400	400	500	600	650	870	1200	1400	1800
Inlet width b1	120	300	400	500	600	800	1000	1300	1600
Constriction width b2	48	120	160	200	240	320	400	520	640
Recess in basin b3	150	330	430	530	630	830	1030	1330	1630
Width of half-shell C	36	90	120	150	180	240	300	390	480



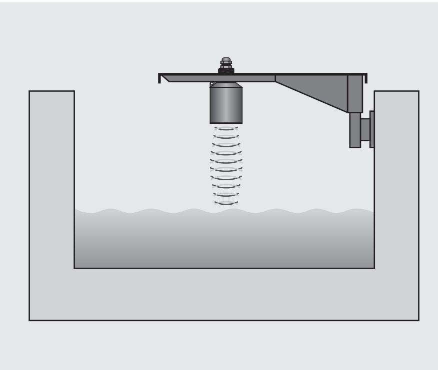


If no Khafagi-Venturi flume or half-shell or weir can be mounted in a channel or a partially-filled pipe, then the Canalflo flow system may be used. More information on the system may be found in the appropriate technical documentation (TI 042/11/e).

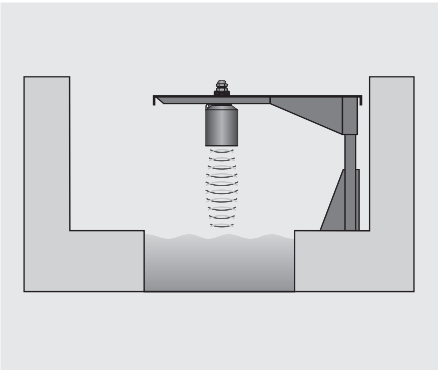
Requirements for Mounting the Sensor

Measuring the upper water level with „Prosonic“ ultrasonic sensor

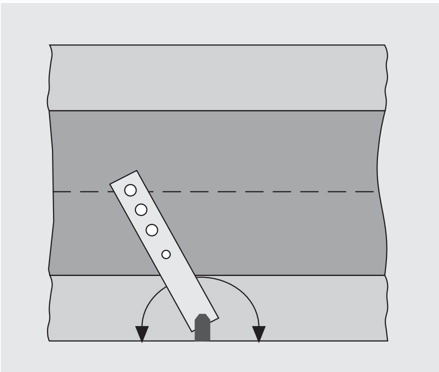
Mount the Prosonic FDU 8.. for measuring the height (upper water level) approx. one channel width b1 upstream from the inlet of the flume.
An ultrasonic sensor which is not in contact with the water is best mounted using a special bracket to position the sensor at a specific distance from the surface of the water and from the channel wall. The sensor surface must always be mounted parallel to the surface of the water.
The Prosonic FMU 861 transmitter can be mounted in different ways either in the field or in the control room.
More detailed information for installation of the sensor and start-up of the transmitter is given in the appropriate Prosonic documentation (BA 100/00/e).



Mounting bracket, wall holder and extension arm



Mounting bracket, stand and extension arm



Mounting bracket can be swivelled to position the sensor over the centre of the flume

Supplementary Documentation

- ❒ Ultrasonic sensor
  Prosonic FDU 80
  Technical Information
  TI 189F/00/e

- ❒ Ultrasonic sensor
  Prosonic FDU 861
  Technical Information
  TI 190F/00/e

- ❒ Prosonic FDU 80/FMU 861
  Operating manual
  BA 100F/00/e

- ❒ Measurement without constrictions
  Canalflo KDR 100/KDS 100/RDS 100
  Technical Information
  TI 042/11/e

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06.97

TI003/11/en/06.97
Printed in FR of Germany